

Contextual Word Sense Disambiguation Using the T5 Transformer Model

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Abstract

1 Introduction

Natural language sentences often contain **ambiguous words whose correct meaning depends on the surrounding context**, making it difficult for automated systems to interpret text accurately. Traditional approaches struggle to capture deep contextual relationships needed to resolve such ambiguities. Therefore, the problem is to **design a system that can automatically determine the correct sense of a word in a given sentence**. In this work, the task is formulated as a text-to-text learning problem using the transformer-based model T5, where the input is a sentence containing an ambiguous word and the model must generate the appropriate meaning or sense based on contextual information. The goal is to improve semantic understanding in natural language processing applications by accurately resolving word sense ambiguity.

2 Objectives

1. Perform exploratory data analysis (EDA) on the WiC dataset to understand the distribution of target words, sentence structures, and label balance.
2. Preprocess and reformat the dataset into a text-to-text format compatible with the T5 architecture.
3. Fine-tune the T5 transformer model to perform the word sense disambiguation task using contextual sentence information.
4. Optimize training through hyperparameter tuning and efficient training strategies to improve model learning and stability.
5. Evaluate the model's ability to correctly determine whether a word has the same meaning across different sentence contexts, improving semantic understanding in NLP systems.
6. Analyze model behavior and identify potential improvements for handling ambiguous words and complex contextual relationships in natural language.

3 Dataset Description

The dataset used in this project is the Word-in-Context (WiC) dataset, which is part of the SuperGLUE benchmark for evaluating natural language understanding systems. The dataset is specifically designed for the word sense disambiguation task.

Each sample in the dataset consists of:

- A target word
- Two sentences containing the same target word
- A label indicating whether the word has the same meaning in both sentences

The labels are represented as:

- 1 (Yes) – The word has the same meaning in both sentences
- 0 (No) – The word has different meanings in the sentences

The labels are represented as 1 (Yes) if the word has the same meaning in both contexts and 0 (No) if the meanings are different. The dataset contains separate training, validation, and test splits, enabling proper model training and evaluation.

The WiC dataset includes both noun and verb usages, which helps evaluate the model's ability to understand contextual semantics across different grammatical forms. The dataset is widely used for benchmarking models that aim to improve contextual understanding in natural language processing.

4 Methodology and Project Plan

Phased implementation following the course timeline (Weeks 1–15), using Python (PyTorch, Hugging Face Transformers, Pandas, Scikit-learn) in Jupyter Notebook / Google Colab with GitHub for version control.

4.1 Phase 1 (Weeks 1–2): Data Preparation

- Environment setup and dataset preparation. Install required libraries and load the WiC dataset from the SuperGLUE benchmark using the Hugging Face datasets library. Perform initial preprocessing, including extracting target words, sentence pairs, and labels, and converting them into a structured format suitable for model training.

4.2 Phase 2 (Weeks 3–4): Model Training

- Exploratory Data Analysis (EDA) to understand dataset characteristics such as word distribution, sentence length, and label balance. Visualize patterns and analyze how ambiguous words appear in different contexts.

4.3 Phase 3 (Weeks 5–7): Domain Dataset Creation

- Data preprocessing and transformation into the text-to-text format required by the T5 model. Construct input sequences containing both sentences and the target word, and generate corresponding output labels (“Yes” or “No”). Perform tokenization using the T5 tokenizer and prepare datasets for training using PyTorch DataLoaders.

4.4 Phase 4 (Weeks 8–11): Baselines and Analysis

- Model training and fine-tuning. Fine-tune the T5 transformer model on the prepared dataset to learn contextual relationships between words and sentences. Apply hyperparameter tuning techniques to optimize training parameters such as learning rate, batch size, and dropout.

4.5 Phase 5 (Weeks 12–15): Final Evaluation

- Model evaluation and analysis. Evaluate the trained model on validation and test datasets using standard evaluation metrics. Analyze model behavior on ambiguous cases and perform error analysis to identify potential improvements in handling contextual ambiguity.

Mid Deadline (March 28): Completion of dataset preprocessing, EDA, and initial model setup.

Final Deadline (April 25): Fully implemented pipeline including preprocessing, model training, evaluation, and analysis.

5 Expected Outcomes

The project will produce:

- Development of a transformer-based system capable of performing word sense disambiguation using contextual information from sentences.
- A fine-tuned T5 model that can determine whether a target word has the same meaning across different sentence contexts.
- Improved understanding of how transformer architectures and self-attention mechanisms capture contextual relationships in text.
- Insights from error analysis that highlight challenging cases of ambiguity and suggest potential improvements for future work.
- A complete NLP pipeline including data preprocessing, model training, evaluation, and analysis for the WSD task.

References

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